



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2023-24
HALF YEARLY EXAMINATION

CLASS: X
SUBJECT: CHEMISTRY [SET A]

FULL MARKS: 80
TIME: 2 HOURS

*Answers to this paper must be written on the paper provided separately.
You will not be allowed to write during first 15 minutes.
This time is to be spent in reading the question paper.
The time given at the head of this Paper is the time allowed for writing the answers.
Section A is compulsory. Attempt any four questions from Section B.
The intended marks for questions or parts of questions are given in brackets [].
(This question paper contains seven printed pages)*

SECTION A

(Attempt all questions from this Section)

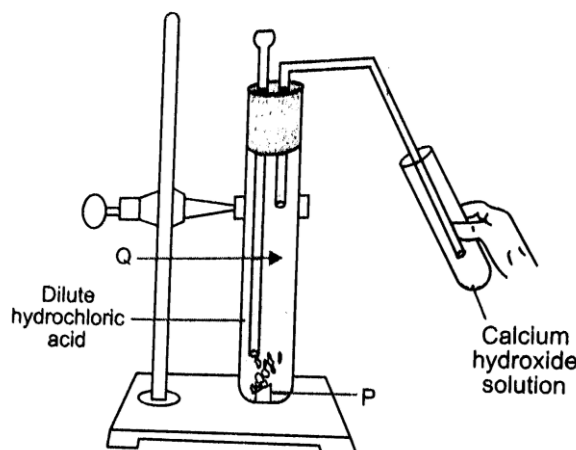
Question1

Choose the correct answers to the questions from the given options.
(Do not copy the question, write the correct answers only)

[15]

- (i) Which of the following sets does not represent empirical formula?
P: C_2H_2O , $C_3H_8O_6$, CHO
Q: CHO , CH_3O , CH_2O_5
R: $C_6H_9O_3$, $C_2H_4O_2$, $C_5H_{10}O_5$
S: CH_2O , CH_3O , CH_5O
- (a) Only P
(b) Only Q
(c) Only R
(d) Both Q and R
- (ii) An element P when burns in oxygen form a basic oxide. The atomic number of P will be?
(a) 7
(b) 9
(c) 14
(d) 11
- (iii) Which of the reactions will need the maximum amount of energy?
P: $N \rightarrow N^{3-} - 3e^-$
Q: $K \rightarrow K^+ + e^-$
R: $K^+ \rightarrow K^{2+} + e^-$
S: $K^+ \rightarrow K - e^-$
- (a) Only P
(b) P and S
(c) Only R
(d) Q and R

- (iv) An experiment was performed in the laboratory which is shown in the figure given below. Study the set-up properly and choose the correct option from the following:



Option	P	Q	Change observed in calcium hydroxide solution
W	K_2CO_3	Cl_2 gas	No change
X	$KHCO_3$	CO_2 gas	No change
Y	$KHCO_3$	H_2 gas	Turns milky
Z	K_2CO_3	CO_2 gas	Turns milky

- (a) W
- (b) X
- (c) Y
- (d) Z

- (v) Few drops of concentrated sulphuric acid to cane sugar to give pure form of carbon. The reaction taking place is:

- (a) Oxidation
- (b) Combustion
- (c) Dehydration
- (d) Combination

- (vi) In the chemistry laboratory four salt solutions [P, Q, R and S] were given to the students and the teacher asked to add excess of ammonium hydroxide solution to it. The students observed that the precipitate formed initially dissolved in excess ammonium hydroxide solution.

P: Zinc sulphate

Q: Aluminium chloride

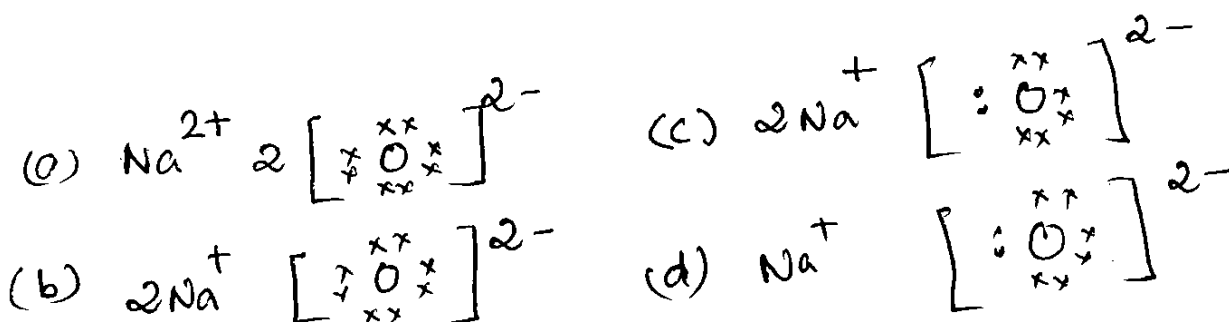
R: Ferric sulphate

S: Ferrous sulphate

The salt solution given by the teacher was:

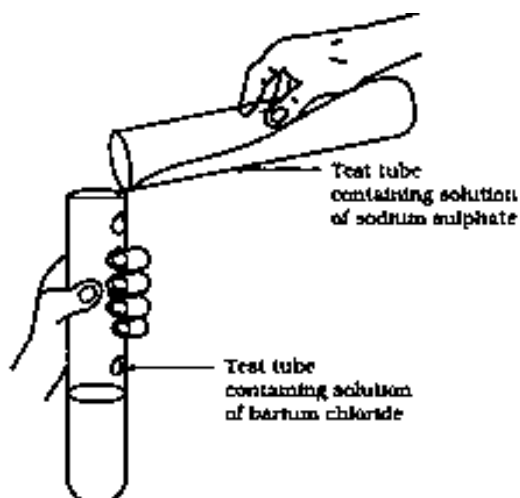
- (a) P
- (b) Q
- (c) R
- (d) S

- (vii) When few drops of methyl orange is added to hydrochloric acid the colour changes to:
 (a) Red
 (b) Pink
 (c) Blue
 (d) Yellow
- (viii) Compound X on hydrolysis yielded a metal hydroxide and ammonia. Here X is ____
 (a) Metal nitrate
 (b) Metal nitride
 (c) Metal nitrite
 (d) Metal azide
- (ix) After performing certain experiments it was observed by the students that hydrogen chloride gas gives white precipitate with:
 (a) Silver nitrate and lead nitrate
 (b) Ammonium nitrate and potassium nitrate
 (c) Sodium nitrate and aluminium nitrate
 (d) (b) and (c) both
- (x) When zinc is treated with dilute nitric acid, the gas formed is ____.
 (a) N_2O_2
 (b) N_2O
 (c) NO_2
 (d) NO
- (xi) Between $NaCl$ and CCl_4 , $NaCl$ has highest melting point because:
 (a) It has maximum ionic character
 (b) It has minimum ionic character
 (c) It has molecules
 (d) It has least molecular weight
- (xii) Which one of the following correctly represents sodium oxide?



- (xiii) An element M has 50% of the electron filled in the 3rd shell as in the 2nd shell.
 The atomic number of M is:
 (a) 10
 (b) 12
 (c) 14
 (d) 18

- (xiv) Study the image below. Identify the product which represents the solid state in the reaction and its colour.

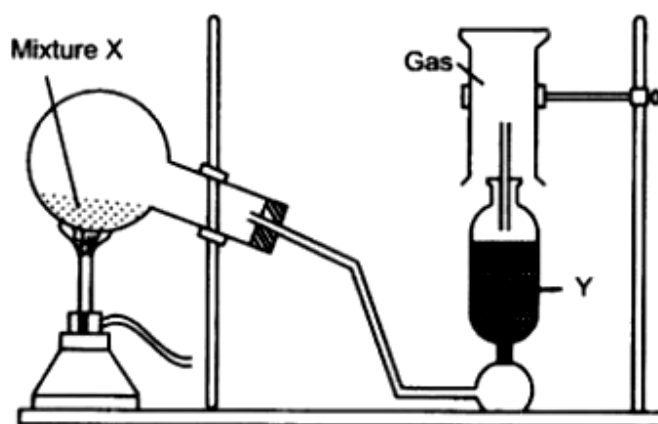


- (a) Barium chloride, white
(b) Barium sulphate, white
(c) Sodium chloride, white
(d) Sodium sulphate, white.
- (xv) Three test tubes A, B and C were taken that contained distilled water, an acidic solution and a basic solution respectively. A red litmus solution was used for testing these solutions, the observed colour changes then would have been:
- (a) A- no change, B- becomes green, C- becomes blue
(b) A- becomes light red, B- becomes blue, C- becomes red
(c) A-remains red, B- no change, C- becomes blue
(d) A- becomes pinkish red, B- becomes dark red, C- becomes blue

Question 2

- (i) The diagram shows the experimental setup of the laboratory preparation of a pungent smelling gas. Study the diagram and answer the following questions:

[5]



- (a) What is the composition of X? Identify Y.
(b) Why is the round bottom flask kept in an inclined position? If the gas is passed through oxygen gas over heated platinum in the combustion tube, what will you observe?
(c) Write a balanced chemical equation for the reaction taking place.

- (ii) Match the following Column A with Column B. [5]

<u>Column A</u>	<u>Column B</u>
(a) Methane	1. Alkaline earth metal
(b) Sulphuric acid	2. Electrovalent
(c) Hydrochloric acid	3. Contact Process
(d) Lithium oxide	4. Dissolution in water
(e) Calcium	5. Covalent

- (iii) Complete the following statements by choosing the correct answer from the bracket. [5]
- (a) When copper strip is dipped in silver nitrate solution colour of the solution turns _____ (grey / blue).
- (b) The basicity of formic acid is ____ (1/2).
- (c) Lead sulphide is prepared by _____ method (synthesis / simple displacement).
- (d) _____ (ammonium nitrate/ ammonium sulphate) is not used in the laboratory preparation of ammonia.
- (e) If an element has seven electrons in the valence shell then it is likely to have the _____ (smallest / largest) atomic size amongst all the elements in the same period.
- (iv) State what do you observe when: [5]
- (a) Ammonium hydroxide solution is added in excess to copper nitrate solution.
- (b) Dilute hydrochloric acid is added to lead nitrate solution and then heated.
- (c) Dilute sulphuric acid is added to sodium sulphide.
- (d) Dilute nitric acid is added to magnesium at very low temperatures.
- (e) Silver nitrate solution is added potassium chloride solution.
- (v) The following table shows the position of six elements A, B, C, D, E and F in the periodic table.

Groups Periods	1	2	3 to 12	13	14	15	16	17	17
2	A					B		C	
3		D			E			F	

- Using the table answer the following: [5]
- (a) Which element will form only covalent compounds?
- (b) Out of D and E which has bigger atomic radius and why?
- (c) Write a common name for the family of elements C and F.
- (d) Write the formula of the compound formed between D and B

SECTION B

(Attempt any four questions)

Question 3

- (i) An element Q is placed in the 2nd group and 3rd period of the Periodic Table. It burns in presence of oxygen to form a basic oxide. [2]
- (a) Identify the element Q.
- (b) Draw the electron dot structure for the formation of this oxide.
- (ii) Write the equation(s) for the reaction(s) to prepare lead sulphate from lead carbonate. [2]

- (iii) Leena prepared hydrochloric acid in the laboratory by passing hydrogen chloride gas in water. [3]
 (a) Draw a diagram to show the arrangement for the absorption of HCl gas in water.
 (b) State why such an arrangement is necessary? Give two reasons for the same.
- (iv) (a) State whether the following statements are TRUE or FALSE. Justify your answer. [3]
 1. Carbon reacts with nitric acid to produce nitrogen dioxide, carbon dioxide and water.
 2. Zinc reacts with dilute nitric acid to liberate hydrogen gas.
 (b) Calculate the number of moles of 60 g of nitric oxide. [N=14, O=16] [3]

Question 4

- (i) How would you distinguish between the following pairs of compounds using a suitable reagent: [2]
 (a) Calcium nitrate and magnesium chloride
 (b) Ferrous sulphate and ferric sulphate.
- (ii) Calculate the vapour density and molecular mass of carbon dioxide if 200mL of the gas at STP weighs 0.40g. (1L of hydrogen at STP weighs 0.09g) [2]
- (iii) Write a balanced chemical equation for the following: [3]
 (a) Ammonia acts as a reducing agent
 (b) Ammonia burns in oxygen
 (c) Excess chlorine passed through ammonia.
- (iv) Write the chemical equations for the industrial preparation of nitric acid. [3]

Question 5

- (i) Ranjana wants to prove the non volatile nature of sulphuric acid. To demonstrate this she adds sulphuric acid to sodium chloride at very low temperature and brings a glass rod dipped in ammonia solution near the mouth of the test tube. [2]
 (a) What will she observe?
 (b) Write a balanced chemical equation for the above reaction.
- (ii) Name the following: [2]
 (a) A triacidic base.
 (b) A hydrogen containing compound which is not an acid
- (iii) Reema takes a blue crystalline salt P in a hard glass test tube. On heating she observed it produces an anhydrous powder Q. Q is dissolved in water. Iron filings are added to one part of the solution and to another part of the solution barium chloride are added. [3]
 (a) Name the compound P.
 (b) Mention what will she observe when iron is added to the solution of Q by mistake.
 (c) State the observation when barium chloride is added to other part of solution Q.
- (iv) The electron affinity of an element X is greater than that of element Y. [3]
 (a) How is the oxidizing power of X likely to compare with that of Y?
 (b) How is the electronegativity of X likely to compare with that of Y?
 (c) State whether X is likely to be placed to the left or to the right of Y in the periodic table?

Question 6

- (i) Name the following: [2]
 (a) An acid which on dilution produces another acid.
 (b) A non metal which on reacting with concentrated nitric acid produces an acid.
- (ii) Write balanced chemical equations for the following conversions [2]
 (a) Sodium nitrate to nitric acid in the laboratory.
 (b) Hydrogen chloride to chlorine using an oxidizing agent.

- (iii) Calculate the following: [3]
 (a) Mass and number of molecules of 50 cc of carbon monoxide at STP.
 (b) Number of moles in 7 gram of nitrogen. [N=14, C=12, O=16]
- (iv) Answer the following questions pertaining to chemical bonding: [3]
 (a) _____ is a covalent molecule having one lone pair of electrons.
 (b) _____ is the positive ion formed when hydrogen chloride is dissolved in water.
 (c) Draw the electron dot diagram of the structure of the above mentioned ion in (b).

Question 7

- (i) A compound consists of 49% carbon, 2.72% hydrogen and 48.3% chlorine by mass. A lab analysis by Anita shows that the compound has a molar mass of 147. Find out the molecular formula. [C=12, O=16, H=1] [3]
- (ii) The distribution of electrons in three elements is as under:
 U = 2, 6; V = 2, 8, 7 and W = 2, 8, 1.
 Use the letter U, V and W as the symbols of elements in answering the following questions. In each case give the formula of molecule or compound formed: [3]
 (a) Between two atoms of U.
 (b) Between U and W.
 (c) Between V and W
- (iii) A mixture of 10cc of CO, 60cc of H₂ and 25 cc of CH₄ are mixed with 750 cc of air (containing 20% oxygen) and ignited. Calculate the composition of the resultant mixture on cooling to room temperature. [4]

Question 8

- (i) State giving reasons if: [2]
 (a) Lead and aluminium can be distinguished by heating the metal powder with concentrated potassium hydroxide solution.
 (b) Hydrogen sulphide and sulphur dioxide can be distinguished by moist lead acetate paper.
- (ii) Name a basic gas that turns Nessler's reagent pale brown. Draw the electron dot diagram of it. [2]
- (iii) Write balanced equations for the following reactions: [3]
 (a) Concentrated sulphuric acid is added to carbon particles.
 (b) Dilute nitric acid is added to copper turnings.
 (c) Warm water is added to aluminium nitride.
- (iv) Identify the following substances: [3]
 (a) On performing the flame test, a salt produces a lilac coloured flame and its solution gives a white precipitate with silver nitrate solution, which is soluble in ammonium hydroxide solution.
 (b) When dilute HCl is added to a salt, a brisk effervescence is produced and the gas turns lime water milky. When NH₄OH solution is added to the above mixture (after adding dilute HCl), it produces a white precipitate which is soluble in excess NH₄OH solution.
 (c) The compound formed when iron and chlorine are heated together.